BASIC: Distributed Task Assignment With Auction Incentive in UAV-Enabled Crowdsensing System

Qichao Xu, Zhou Su, *Senior Member, IEEE*, Dongfeng Fang, *Member, IEEE*, and Yuan Wu

***Abstract*—With the wide adoption of unmanned aerial vehicles (UAVs) in our daily life, using UAVs to collect sensing data from the surrounding environment has become a practice. Notably, people use UAVs to capture videos and upload them to Youtube. Accordingly, using distributed UAVs in the city to collect real-time data represents a key enabling approach for smart city, e.g., air pollution and traffic monitoring. To achieve this goal, in this work, we develop a UAV-based crowdsensing system that allows UAVs to dynamically join and contribute to data collection. We make three contributions. First, due to UAVs' highspeed spatial mobilities, diverse sensing capabilities, and limited energy reserves, it is key to stimulate UAVs to cooperatively perform sensing tasks. To address this issue, we develop a game-theory-based distributed UAV incentive scheme, named BASIC. Second, to make BASIC work fully distributed and scalable, we first develop a mathematical model to characterize the interdependent operations of task requesters and collaboratively data sensing. We then devise a centralized auction mechanism with truthful bidding for UAVs. A heuristic algorithm is then devised to select the winning UAVs for sensing tasks and determine the proper payments to winners. Last, we perform extensive simulations to validate the effectiveness of proposed scheme and show that BASIC can provide effective incentive as compared to the existing schemes.**

***Index Terms*—UAV-enabled crowdsensing system, incentive, distributed task assignment, auction mechanism.**

I. Introduction

RECENTLY, unmanned aerial vehicle (UAV) as a promising intelligent device has become integral to our life. During the outbreak of COVID-19, a massive of UAVs are utilized to disseminate information of epidemic prevention and perform disinfection tasks in residential areas [1], [2], [3], [4]. It is reported that the global shipment of enterprise UAVs is expected to total 1.3 million units by 2023, which is the fourfold of those in 2019 [5]. In a series of UAVs' applications, UAV-enabled crowdsensing attracts much attention from academia and industry [6], [7], [8], [9], [10], where UAVs equipped with powerful built-in sensors (e.g., thermometer, hygrometer, air particulates measure meter, etc.) are utilized to timely collect information of the surrounding environment. For example, UAVs equipped with cameras can be employed to detect whether there exist crowd gathering phenomena in a residential area during the outbreak of an infection [11], [12].

Compared with the existing crowdsensing system [8], [13], [14], [15], [16], [17], [18], [19] that employs mobile users to achieve information perception, UAV-enabled crowdsensing presents many differentiations. On one hand, as the number of mobile users' mobile devices is small, the types of sensors on the mobile devices are limited and their sensing qualities are usually poor, while UAVs can be equipped with multiple kinds of high-precision sensors to perform sophisticated sensing missions. On the other hand, mobile devices can only carry out small indoor region sensing with limited mobility patterns, but UAVs with flexible spatial mobility are able to fly to farther places where mobile users cannot reach in time, and collect multidimensional sensing data including ground perception information and air sensory information that mobile devices are hard to collect.

However, since it is costly for UAVs to execute sensing tasks, the owners of UAVs exhibit a certain selfishness, where not only limited resources (e.g., bandwidth, computing, energy, etc.) of UAVs are cost, but also the time of UAVs owners is also consumed. As such, without an appropriate incentive mechanism, UAVs are reluctant to participate in sensing data collection, since they cannot receive sufficient payments from task requesters to compensate for their sensing costs. Besides, due to the features of UAVs including diverse types of sensors, flexible spatial mobilities, and limited battery capacities, the multidimensional sensing capacities and the personalized compensation requirements of UAVs should be taken into full consideration in the incentive mechanism. Therefore, it is urgent to devise an incentive-based sensing task assignment scheme for improving the performance of UAV-enabled crowdsensing.

Manuscript received 19 June 2022; revised 7 January 2023 and 14 April 2023; accepted 24 April 2023. Date of publication 2 August 2023; date of current version 13 February 2024. This work was supported in part by the National Natural Science Foundation of China under Grants 62001278, 62273223, and U20A20175, in part by the Natural Science Foundation of Shanghai under Grant 20ZR1420700, in part by the National Key R&D Program of China under Grant 2022YFB3104500, in part by Shanghai Sailing Program under Grant 20YF1413600, in part by the Science and Technology Development Fund of Macau SAR under Grant 0162/2019/A3, and in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2022A1515011287. The review of this article was coordinated by Prof. Linke Guo. *(Corresponding author: Zhou Su.)*

Qichao Xu is with the School of Mechatronic Engineering and Automation, Shanghai University, Shanghai 200444, China (e-mail: xqc690926910@shu.edu.cn).

Zhou Su is with the School of Cyber Science and Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: zhousu@ieee.org).

Dongfeng Fang is with the Department of Computer Science, California Polytechnic State University, San Luis Obispo, CA 93407 USA (e-mail: dofang@calpoly.edu).

Yuan Wu is with the Department of Computer and Information Science, University of Macau, Macau, China (e-mail: yuanwu@um.edu.mo).

Digital Object Identifier 10.1109/TVT.2023.3299428

0018-9545 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

Although a number of works [20], [21], [22], [23], [24], [25], [26], [27], [28] have studied the incentive of cooperative perception in the crowdsensing, they cannot be directly applied in UAV-enabled crowdsensing. First, most of the existing works concentrate on the ground data collection with mobile phones, but the utilization of UAVs to collect the air information is not considered. Besides, the sensing cost is the prime concern in most of the existing works, whereas the diverse sensing capabilities of UAVs are not sufficiently taken into account. Additionally, the homogeneous sensing tasks are mainly considered in the existing works, while how to distributedly assign sensing tasks with differentiated requirements is not further addressed. Consequently, it is still an open and vital issue to design a novel incentive scheme for distributed UAVs to cooperatively perform sensing tasks.

We present a reverse-auction-based distriButed tAsk asSIgnment with auCtion UAV incentive scheme, named BASIC, in the UAV-enabled crowdsensing system. Specifically, we first develop a UAV-enabled crowdsensing framework including the cooperative operations of task requesters, a sensing platform, and multiple distributed UAVs. Furthermore, to incentivize the cooperation of distributed UAVs, we develop a reverse-auction-based incentive mechanism for the sensing platform to outsource the sensing tasks. Then, to maximize the utility of the sensing platform and guarantee the positive payoffs for UAVs, we devise a heuristic algorithm to realize the optimal UAV selection and payment determination for the sensing platform. Finally, extensive simulation results show that BASIC has higher utility of the sensing platform and sensing qualities for tasks than those of the conventional schemes. The main contributions of this article are three-fold.

- *System:* We develop a UAV-enabled crowdsensing system, which can be utilized to conduct multidimensional and spatial sensing tasks. In the system, the UAV equipped with multiple-type sensors can be employed to perform multiple sensing tasks simultaneously. Specifically, task requesters first submit sensing tasks with diverse demands to the sensing platform, and the sensing platform then assigns multiple-type sensing tasks to distributed UAVs that meet the demands of the tasks. After collecting and returning the sensing data, UAVs obtain the corresponding payments from the task requesters via the sensing platform.
- *Mechanism:* We propose a reverse-auction-based incentive mechanism to stimulate UAVs for cooperatively participating in task sensing. The utility of the UAV is determined based on payment and cost, where the cost includes energy cost, sensing cost, and transmission cost. The utility of the sensing platform is designed based on the sensing qualities of tasks, task pending time, and payments to UAVs. A problem is formulated to maximize the utility of the sensing platform, considering the diverse demands of sensing tasks and differentiated sensing capabilities of UAVs.
- *Algorithm:* We propose a heuristic algorithm to solve the utility maximization problem for the sensing platform. The heuristic algorithm includes the winning UAV selection sub-algorithm and payment determination sub-algorithm. Specifically, UAVs that provide the highest contributions for the utility of the sensing platform are selected as the winners to perform sensing tasks. The optimal payments are determined based on the critical payment, which can encourage UAVs to submit costs as bids. As such, the truthfulness and individual rationality of BASIC are guaranteed.

The organization of the article is as follows. The related works are reviewed in Section II. Section III proposes the system model. Section IV proposes a UAV-enabled crowdsensing framework and a reverse auction-based incentive mechanism. Section V introduces the problem solution and property analysis. Section VI conducts the extensive simulations to evaluate BASIC and the conclusions are in Section VII.

II. Related Work

In this section, we review the related works including UAV-enabled crowdsensing system, task assignment and incentive in crowdsensing system.

A. UAV-Enabled Crowdsensing System

Due to the flexibility and sensing capability of UAVs, the UAV-enabled crowdsensing system attracts much attention from academia and industry. Dai et al. [29] utilized decentralized deep reinforcement learning to study the delay-sensitive and energy-efficient UAV crowdsensing, for simultaneously maximizing collected data amount, and minimizing the energy cost of all UAVs. Wei et al. [30] proposed a high performance UAV crowdsourcing system with deep reinforcement learning integrated with partial observations, where the actor-critic technique incorporating the attention mechanism is utilized for collaborative sensing data collection. Liu et al. [31] designed a deep learning-based framework to enable a group of UAVs to cooperatively collect data from low-level sensors in an energy-efficient manner, where multiple charging stations are randomly deployed to recharge the batteries. Cai et al. [32] studied the collaborative data sensing with computing offloading in the UAV-assisted crowdsensing system, where the multi-agent deep reinforcement learning is leveraged to maximize the overall system utility. Kang et al. [33] investigated the energy-efficient route planning problem in the UAV-assisted crowdsensing system, where UAVs deployed by different users compete with each other for the limited sensing resources. Most of these works concentrate on the sensing data by optimizing the route planning of UAVs, while the distributed sensing task assignment lacks in-depth and unambiguous studying.

B. Task Assignment in Crowdsensing System

The sensing task assignment has been extensively studied to improve the performance of the crowdsensing system. Tan et al. [34] proposed a group-oriented cooperative approach to realize sensing task assignment in the social mobile crowdsensing system, where the real-life relationship among mobile users in the social network is exploited to improve the task coverage while achieving good task cooperation quality. Wang et al. [35] presented an efficient sensing task assignment in an online location-aware crowdsensing system, which can gradually

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

TABLE I
VARIABLES

Notations	Description
$\mathcal{I}$	The set of UAVs.
$(x_i[t], y_i[t])$	Horizontal location of UAV i at time slot t.
$h_i[t]$	Altitude of UAV i at time slot t.
$v_i[t]$	Flying velocity of UAV i.
V_i^{max}	The maximum allowable flying velocity of UAV i.
E_i^{max}	The maximum energy reserve of UAV i.
$e_i[t]$	Remaining energy of UAV i at time slot t.
P_i^h	Hovering power of UAV i.
$P_i^f[t]$	Flying power of UAV i at time slot t.
$\mathcal{K}$	The set of sensor types.
$\alpha_{i,k} \in \{0,1\}$	Whether UAV i is equipped with the sensor of type k.
Λ	The allowable maximum number of sensors on each UAV.
$q_{i,k} \in [0,1]$	Sensing quality of UAV i on sensor of type k.
$P_{i,k}^s$	Sensing power of UAV i on sensor of type k.
η_i	Conversion parameter between flying velocity and flying power of UAV i.
ν	Positive flying power exponent.
β_0	Channel power gain at the unit reference distance.
μ	Path loss exponent of LoS link.
$d_{i,0}[t]$	Euclidean distance between UAV i and its closest BS at time slot t.
$R_i[t]$	Communication data rate between UAV i and its closest BS at time slot t.
B_i	Transmission bandwidth of UAV i.
P_i^t	Transmission power of UAV i.
σ_0^2	Noise power spectrum density.
$\mathcal{M}$	The set of sensing tasks.
Γ_m	The six-tuple of sensing task m.
(X_m, Y_m)	Horizontal location requested by sensing task m.
H_m	Altitude required by sensing task m
Φ_m	Sensing type of task m.
Q_m	Sensing quality of task m.
T_m	Sensing duration of task m.
Θ_m	TTL of sensing task m.

gather high-quality sensory data considering location awareness and inferred worker expertise. Ji et al. [36] designed a dynamic task assignment model for crowdsensing with consideration of mobile user availability and tasks changing over time, in which the sensing ability of mobile users is comprehensively evaluated to collect high-quality data for sensing tasks in the target area. Qian et al. [37] presented an optimal location privacy-preserving and service quality guarantee tasks assignment scheme in vehicle-based crowdsensing system, where the task allocation problem is framed with the goal of concurrently minimizing the distance traveled and maximizing the quality of task completion. Li et al. [38] investigated a multi-task assignment scheme with time constraints, where the impact of time constraints on multi-task assignment is analyzed and the utility of the sensing platform is maximized by a novel devised evolutionary algorithm. Although many schemes have been proposed to realize the optimal sensing task assignment in the crowdsensing system, the differentiated and diverse demands of sensing tasks should be further taken into account.

C. Incentive in Crowdsensing System

Many efforts have been made on the incentive for the crowdsensing system. Li et al. [39] utilized the three-stage Stackelberg game to design an incentive mechanism in the continuous and time-varying cases, where sensing tasks are generated by different requesters who pay the platform in each time period.

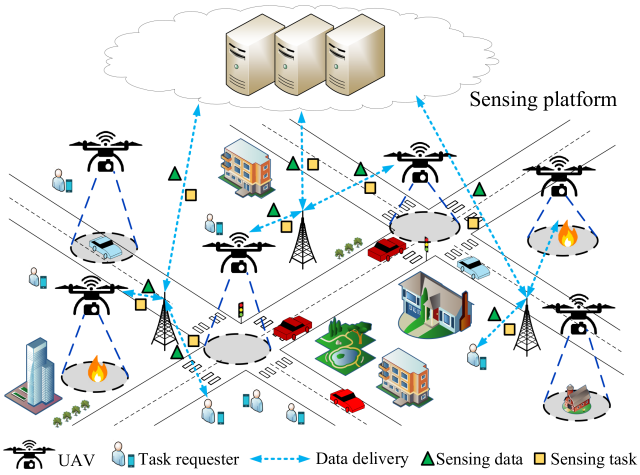


Fig. 1. Illustration of UAV-enabled crowdsensing system.

Lin et al. [40] proposed a multiple round monetary incentive mechanism by considering multi-round cooperation and alternating between task information diffusion and task assignment operations in a mobile crowdsensing system. Nie et al. [41] proposed an incentive scheme for multiple crowdsensing service providers and multiple users, in which a Stackelberg game with multiple leaders and multiple followers is used to model the strategic interactions between service providers and users. Xu et al. [42] studied the incentive problem in a novel and practical crowdsensing scenario with auction theory, where the tasks are related to time windows and the platform has strong demands for data integrity. Ji et al. [43] devised a quality-aware online task incentive mechanism for a crowdsensing system, where the arrival of workers is uneven and the quality of tasks grows in line with workers' efforts. Although many existing works have proposed incentive mechanisms to enhance the performance of the crowdsensing system, few of them consider the effects of UAVs' diverse sensing capacities.

III. System Model

In this section, we introduce the system model including the network model, communication model, task model, and design goals.

A. Network Model

As shown in Fig. 1, a typical UAV-enabled crowdsensing system is considered, which consists of task requesters, a sensing platform, and multiple distributed UAVs. The task requesters are users or organizations that submit tasks to the sensing platform to request sensing data. The sensing platform is in charge of the crowdsensing workflow, which receives the sensing tasks from task requesters, assigns tasks to appropriate UAVs, collects sensing data from them, and finally returns the sensing data to task requesters. The UAVs are performers of sensing tasks, who collect sensing data for requesters at designated points of interest.

Let $\mathcal{I} = \{1, 2, \ldots, i, \ldots, I\}$ denote the set of UAVs. The horizontal location of UAV i at time slot t is denoted as $(x_i[t], y_i[t])$.

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

The altitude of UAV i at time slot t is $h_i[t]$. The flying velocity is constant for each UAV. The constant flying velocity of UAV i in meter per second (m/s) is denoted as v_i. Since the sizes of UAVs are generally limited and they cannot be equipped with large energy storage devices, the energy of each UAV is constrained and it needs to be recharged every some time slots. Let E_i^{max} denote the maximum energy reserve of UAV i. The remaining energy of UAV i at time slot t is denoted as $e_i[t]$, whereby $e_i[t] \leq E_i^{max}$. The hovering power of UAV i is denoted as P_i^h. The flying power is related to the flying velocity. Generally, the flying power is in direct proportion to the flying velocity, i.e., a high flying velocity requires a large flying power. For simplicity, the flying power of UAV i at time slot t can be given by [44]

$$P_i^f = \eta_i (v_i)^\nu, \tag{1}$$

where η_i is the conversion parameter between flying velocity and flying power of UAV i, which is related to the weight of UAV and air density, etc [44]. ν is a positive flying power exponent.

Each UAV is equipped with multiple types of sensors to perceive its surrounding environment, e.g., thermometer, hygrometer, infrared camera, etc. Assume that there exist K types of sensors in the system, the set of which is denoted as $\mathcal{K} = \{1, 2, \ldots, k, \ldots, K\}$. We set a binary variable for each UAV to indicate whether it is equipped with a designated sensor. Specifically, $\alpha_{i,k} = 1$ means that the sensor of type k is mounted on UAV i. Otherwise, $\alpha_{i,k} = 0$. It is notable that the number of sensors mounted on each UAV is restricted, due to its small size and limited energy, i.e., $\sum_{k=1}^{K} \alpha_{i,k} \leq \Lambda, \forall i \in \mathcal{I}$. Here Λ is the allowable maximum number of sensors mounted on each UAV. Additionally, UAVs have different sensing qualities on the same type of sensor, which can be evaluated based on the sensing precisions. Let $q_{i,k}$, $q_{i,k} \in [0, 1]$ denote the sensing quality of UAV i on sensor of type k. Here, $q_{i,k} = 0$ means that UAV i does not have sensing quality on the sensor of type k, which equals to that UAV i is not equipped with the sensor of type k, while $q_{i,k} = 1$ indicates that UAV i has the highest sensing quality on sensor of type k. Besides, it needs a certain power for a UAV to execute sensing tasks. The sensing power of UAV i on sensor of type k is denoted as $P_{i,k}^s$.

B. Communication Model

In the system, UAVs submit sensing data to the sensing platform via wireless links to their closest base stations (BSs). According to the channel measurements, the UAV-to-ground channel is usually composed of a strong line-of-sight (LoS) link [45]. Therefore, for simplicity, the channel between each UAV and its nearest BS is assumed to be modeled by a LoS link. As such, the channel power gain from UAV i to its closest BS at time slot t is given by [46]

$$\beta_i[t] = \beta_0 (d_{i,0}[t])^{-\mu}, \tag{2}$$

where β_0 denotes the UAV-to-ground channel power gain at the unit reference distance, μ is the path loss exponent of the LoS link, and $d_{i,0}[t]$ is the Euclidean distance between UAV i and its closest BS at time slot t. Then, according to Shannon-Hartley theorem [47], the communication data rate between UAV i and its closest BS at time slot t can be obtained by

$$R_i[t] = B_i \log_2 \left(1 + \frac{P_i^t \beta_0 (d_{i,0}[t])^{-\mu}}{\sigma_0^2 B_i}\right), \tag{3}$$

where B_i is the transmission bandwidth of UAV i, P_i^t is the transmission power of UAV i, and σ_0^2 is the noise power spectrum density.

C. Task Model

In the system, task requesters submit sensing tasks to the sensing platform over time. Assume that there exist M sensing tasks that are hosted by the platform at time slot t, whose set is $\mathcal{M} = \{1, 2, \ldots, m, \ldots, M\}$. As generated from different task requesters, sensing tasks have differentiated sensing demands including horizontal location, altitude, type, sensing quality, sensing duration, and time-to-live (TTL). As such, we utilize a six-tuple to describe sensing task m, which is denoted as $\Gamma_m = \langle (X_m, Y_m), H_m, \Phi_m, Q_m, T_m, \Theta_m \rangle$. Here, (X_m, Y_m) and H_m are the horizontal location and altitude required by sensing task m, respectively. Φ_m ($\Phi_m \in \mathcal{K}$), Q_m, T_m, and Θ_m are type, sensing quality, sensing duration, and TTL of sensing task m, respectively. Without loss of generality, a sensing task can be assigned to at most one UAV. Meanwhile, each UAV can perform at most one sensing task at each moment, as sensing tasks require to perceive data at different locations.

D. Design Goals

In this article, BASIC has two design goals. One is to motivate UAVs to cooperatively participate in crowdsensing. The other is to distributedly assign sensing tasks to the optimal UAVs that provide the maximum sensing qualities, the shortest sensing executing time, and the minimum sensing costs.

IV. UAV-Enabled Crowdsensing Framework and Incentive Mechanism

In this section, to achieve the goals of BASIC, we first develop a UAV-enabled crowdsensing framework. Then, to motivate the cooperation of UAVs, we devise a reverse-auction-based incentive mechanism, followed by the problem formulation for BASIC.

A. UAV-Enabled Crowdsensing Framework

As shown in Fig. 2, the UAV-enabled crowdsensing framework includes interdependent operations of task requesters, sensing platform, and distributed UAVs, which are summarized as follows.

- First, task requesters submit sensing tasks to the sensing platform. The sensing platform then aggregates the sensing tasks and publishes them to distributed UAVs with the description of each sensing task. (Step ①)
- Then, a reverse-auction-based incentive mechanism is started by the sensing platform to motivate UAVs to participate in crowdsensing. Each UAV selects its interested sensing tasks based on the requirements of sensing tasks

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

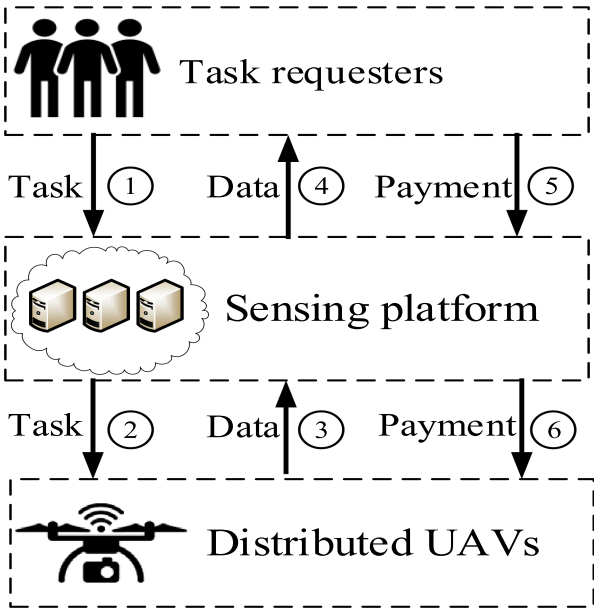


Fig. 2. UAV-enabled crowdsensing framework.

and determines its optimal bid with the sensing cost. According to UAVs' bids and their sensing capabilities, the sensing platform selects winning UAVs and assigns the sensing tasks to them. (Step ②)

- Afterwards, each UAV flies to the designated spatial location to collect data according to the requirement of the assigned sensing task, i.e., sensing type, sensing quality, sensing duration, and TTL. After completing the sensing tasks, UAVs submit sensing data to the sensing platform which then returns sensing data to the corresponding task requesters. (Steps ③∼④)
- Finally, the task requesters pay for the services of the sensing platform and each winning UAV receives the corresponding payments from the sensing platform. (Steps ⑤∼⑥)

For the sake of easy exposition, the task assignment of one time slot is analyzed to determine the optimal decisions of both the sensing platform and UAVs. Indeed, the number of UAVs and sensing tasks changes over time. Besides, the states of UAVs also change with time, including locations, remaining energy, etc. At each time slot, according to the new situations of both UAVs and tasks, the sensing platform re-allocates the corresponding sensing tasks to available UAVs.

B. Reverse-Auction-Based Incentive Mechanism

Performing sensing tasks not only costs the limited resources (e.g., bandwidth, energy, computing, etc.) of UAVs, but also consumes much time of UAVs' owners. As such, the incentive mechanism is needed to stimulate the participation of UAVs. We devise a reverse-auction-based incentive mechanism for motivating UAVs to perform sensing tasks. Specifically, the sensing platform first broadcasts the description information of sensing tasks to all UAVs, each UAV then determines its optimal bid according to the cost of performing the sensing task. After the determination of the optimal bid, each UAV submits a bid to the sensing platform. Then, the sensing platform selects the winning UAVs that jointly provide the maximum utility for itself and determines the payments based on the bids of the winners. Finally, each winning UAV obtains the payment from the sensing platform.

Auction means that multiple buyers go to bid on the same goods through estimations, where the buyer with the highest bidding price is the winner to obtain the goods [48]. In this article, the task requester is the buyer who purchases sensing services from multiple UAVs. The UAVs are sellers to provide different bids by considering the sensing costs, where the UAV with the lowest bidding price is the winner to execute the corresponding sensing task.

1) Utility of UAV: Let $\mathbf{b}_i = (b_{i,1}, b_{i,2}, \ldots, b_{i,m}, \ldots, b_{i,M})$ denote the bid of UAV i, which is a vector to show the bids for M sensing tasks. Specifically, $b_{i,m}$ is the bid of UAV i for sensing task m. The bid is a real number that is larger than or equal to 0, i.e., $b_{i,m} \geq 0$. Especially, if $b_{i,m} = \infty$, it indicates that UAV i is not willing to execute sensing task m. The utility of UAV i to perform sensing task m is defined as

$$U_{i,m}(b_{i,m}) = \begin{cases} p_i - C_{i,m}, & \text{if } z_{i,m} = 1 \\ 0, & \text{otherwise} \end{cases}, \tag{4}$$

where p_i is the payment to UAV i from the sensing platform, $z_{i,m} = 1$ indicates that UAV i is the winner to perform sensing task m, and $C_{i,m}$ is the cost of UAV i to perform sensing task m.

The cost of UAVs to perform sensing tasks includes three parts: energy cost, sensing cost, and transmission cost. Energy cost consists of flying energy cost, hovering energy cost, sensing energy cost, and transmission energy cost. The flying energy of UAV i can be given by

$$E^f_{i,m} = P^f_i \cdot \frac{d_{i,m}[t]}{v_i} = \eta_i \left(v_i\right)^{\nu-1} d_{i,m}[t], \tag{5}$$

where $d_{i,m}[t]$ is the Euclidean distance between the spatial location of UAV i at time slot t and the spatial location required by sensing task m. It can be expressed by [49]

$$d_{i,m}[t] = \sqrt{\left(x_i[t] - X_m\right)^2 + \left(y_i[t] - y_m\right)^2 + \left(h_i[t] - H_m\right)^2}. \tag{6}$$

According to the hovering power and sensing duration, the hovering energy of UAV i for sensing task m can be obtained by $E^h_i = P^h_i T_m$. Similarly, the sensing energy of UAV i for sensing task m is $E^s_i = P^s_{i,\Phi_m} T_m$, where P^s_{i,Φ_m} is the sensing power of UAV i for task m. After executing sensing tasks, UAVs upload the sensing results to the sensing platform via the communication link to the BS, which induces a certain energy consumption. As such, the transmission energy of UAV i for uploading sensing data of task m can be obtained by

$$E^t_{i,m} = P^t_i \frac{D_{i,m}}{R_i\left[t_{i,m}\right]}, \tag{7}$$

where $D_{i,m}$ is the data size of sensing result for sensing task m, which can be estimated by UAV i based on its historical sensing experience with some learning methods [50]. $R_i[t_{i,m}]$

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

and $d_{i,0}[t_{i,m}]$ are the transmission rate of UAV i and the Euclidean distance between the UAV i and its closest BS after UAV i completes sensing task m, respectively. After completing a sensing task, the UAV flies to the closest BS and uploads the sensing result to the sensing platform. For example, when UAV i completes sensing task m, the time length of UAV i flying to the closest BS is $d_{i,m}[t]/v_i$. $t_{i,m}$ is the time slot when UAV i completes sensing task m. It is obtained by

$$t_{i,m} = t + T_m + \frac{d_{i,m}[t]}{v_i}. \tag{8}$$

Aggregating the flying energy, hovering energy, sensing energy, and transmission energy, the energy consumption of UAV i to execute sensing task m is

$$\begin{aligned} E_{i,m} &= E^f_{i,m} + E^h_{i,m} + E^s_{i,m} + E^t_{i,m} \\ &= \eta_i \left(v_i\right)^{\nu-1} d_{i,m}[t] + P^h_i T_m + P^s_{i,\Phi_m} T_m + \frac{P^t_i D_{i,m}}{R_i\left[t_{i,m}\right]}. \end{aligned} \tag{9}$$

As such, the energy cost of UAV i to perform sensing task m is $C^e_{i,m} = c^e_i E_{i,m}$, where c^e_i is the unit energy cost of UAV i.

Sensing cost means the cost on the operations of sensors to collect data, such as the cost of computing and storage, etc. Notably, the types of sensors have significant effects on the sensing cost. For example, the cost of a thermoscope is much lower than that of an infrared camera. Apart from the sensor type, the sensing quality also affects the sensing costs of UAVs, where high sensing quality induces a large sensing cost. Besides, the sensing cost is in direct proportion to the sensing duration of the task. Therefore, the sensing cost of UAV i to perform sensing task m can be expressed as

$$C^s_{i,m} = c^s_{i,\Phi_m} q_{i,\Phi_m} T_m, \tag{10}$$

where c^s_{i,Φ_m} is the cost of UAV i on the sensor of type Φ_m per unit sensing time with unit sensing quality, q_{i,Φ_m} is the sensing quality of UAV i on the sensor of type Φ_m.

Transmission cost means the cost of UAVs to submit sensing data via wireless links. Two factors affect the transmission cost. On one hand, if the bandwidth between the UAV to its closest BS is large, the cost to purchase the spectrum bandwidth is high. On the other hand, the large transmission time induces a high transmission cost. As such, the transmission cost can be obtained by

$$\begin{aligned} C^t_{i,m} &= c^t_i B_i \cdot \frac{D_{i,m}}{B_i \log_2\left(1 + \frac{P^t_i \beta_0 (d_{i,0}[t_{i,m}])^{-\mu}}{\sigma^2_0 B_i}\right)} \\ &= c^t_i \frac{D_{i,m}}{\log_2\left(1 + \frac{P^t_i \beta_0 (d_{i,0}[t_{i,m}])^{-\mu}}{\sigma^2_0 B_i}\right)}, \end{aligned} \tag{11}$$

where c^t_i is the transmission cost of UAV i per unit bandwidth and unit transmission time.

Combining the energy cost, sensing cost, and transmission cost, the cost of UAV i to perform sensing task m is

$$\begin{aligned} C_{i,m} &= C^e_{i,m} + C^s_{i,m} + C^t_{i,m} \\ &= c^e_i E_{i,m} + c^s_{i,\Phi_m} q_{i,\Phi_m} T_m \\ &\quad + c^t_i \frac{D_{i,m}}{\log_2\left(1 + \frac{P^t_i \beta_0 (d_{i,0}[t_{i,m}])^{-\mu}}{\sigma^2_0 B_i}\right)}. \end{aligned} \tag{12}$$

2) Utility of Sensing Platform: After assigning sensing tasks to UAVs, the sensing platform obtains a certain utility. The utility of the sensing platform can be expressed as

$$U_0(\mathbf{Z}) = \lambda F_0(\mathbf{Z}) - (1-\lambda)\omega C_0(\mathbf{Z}), \tag{13}$$

where $F_0(\mathbf{Z})$ is the satisfaction function from winning UAVs performing sensing tasks, $C_0(\mathbf{Z})$ is the cost function of the sensing platform, λ is the weighted parameter of the satisfaction function, and ω is the adjusting coefficient to balance the satisfaction and payment. Indeed, in different scenarios, the sensing platform has different sensing demands. When the sensing platform pursues high sensing quality, it can set a large weighted parameter λ. When the sensing platform wants to reduce the payment, it can set a small weighted parameter λ. Here, $\mathbf{Z} = [z_{i,m}]_{I\times M}$ is the winning UAV selection matrix. Here, $z_{i,m}$ is a binary variable, where $z_{i,m} = 1$ implies UAV i is selected as a winner to perform sensing task m and otherwise $z_{i,m} = 0$.

$F_0(\mathbf{Z})$ is related to the sensing quality and task pending time. A higher sensing quality generally makes the sensing platform obtain more precise sensing data. The task pending time (i.e., Π_0) includes the time length of the winning UAV flying to the spatial location required by the sensing task and the delay of transmitting sensing data to the sensing platform. It is obtained by

$$\Pi_0 = \sum_{m=1}^{M}\sum_{i=1}^{I} z_{i,m}\left(\frac{d_{i,m}[t]}{v_i} + \frac{D_{i,m}}{R_i\left[t_{i,m}\right]}\right). \tag{14}$$

As such, the satisfaction function of the sensing platform is expressed as

$$\begin{aligned} F_0(\mathbf{Z}) &= \gamma \sum_{m=1}^{M}\sum_{i=1}^{I} z_{i,m}\varpi q_{i,\Phi_m} - (1-\gamma)\Pi_0 \\ &= \gamma \sum_{m=1}^{M}\sum_{i=1}^{I} z_{i,m}\varpi q_{i,\Phi_m} - (1-\gamma) \\ &\quad \times \sum_{m=1}^{M}\sum_{i=1}^{I} z_{i,m}\left(\frac{d_{i,m}[t]}{v_i} + \frac{D_{i,m}}{R_i\left[t_{i,m}\right]}\right), \end{aligned} \tag{15}$$

where γ is the weighted parameter, and ϖ is the adjusting coefficient to balance the sensing quality and the task pending time. The cost function of the sensing platform is the payments to all winning UAVs, which can be expressed as

$$C_0\left(\mathbf{Z}\right) = \sum_{m=1}^{M}\sum_{i=1}^{I} z_{i,m} p_i. \tag{16}$$

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

Therefore, according to the utility expression defined in (13), the explicit utility function of the sensing platform is obtained by

$$U_0(\mathbf{Z}) = \lambda \left\{ \gamma \sum_{m=1}^{M} \sum_{i=1}^{I} z_{i,m} \varpi q_{i,\Phi_m} - (1-\gamma) \times \sum_{m=1}^{M} \sum_{i=1}^{I} z_{i,m} \left(\frac{d_{i,m}[t]}{v_i} + \frac{D_{i,m}}{R_i\left[t_{i,m}\right]} \right) \right\} - (1-\lambda)\omega \sum_{m=1}^{M} \sum_{i=1}^{I} z_{i,m} p_i. \tag{17}$$

C. Problem Formulation

In the reverse-auction-based incentive mechanism, the target of the sensing platform is to select winning UAVs that provide the maximum utility for the sensing platform and determine the proper payments to winners. As such, a maximization problem for the sensing platform is formalized as

Problem 1 :

$$\max\ \lambda \left\{ \gamma \sum_{m=1}^{M} \sum_{i=1}^{I} z_{i,m} \varpi q_{i,\Phi_m} - (1-\gamma) \times \sum_{m=1}^{M} \sum_{i=1}^{I} z_{i,m} \left(\frac{d_{i,m}[t]}{v_i} + \frac{D_{i,m}}{R_i\left[t_{i,m}\right]} \right) \right\} -(1-\lambda)\omega \sum_{m=1}^{M} \sum_{i=1}^{I} z_{i,m}(b_{i,m} + a_{i,m}), \tag{18}$$

$$\text{s.t.} \begin{cases} z_{i,m}\left(E_{i,m} + \eta_i (v_i)^{\nu-1} d_{i,m,0}\right) \le e_i[t], & \text{(19a)} \\ z_{i,m}\left(\frac{d_{i,m}[t]}{v_i} + T_m + \frac{D_{i,m}}{R_i[t_{i,m}]} + t\right) \le \Theta_m, & \text{(19b)} \\ Q_m \le z_{i,m} q_{i,\Phi_m}, & \text{(19c)} \\ z_{i,m}\alpha_{i,\Phi_m} + z_{i,m} \in \{0,2\}, \forall i \in \mathcal{I}, \forall m \in \mathcal{M}, & \text{(19d)} \\ \sum_{i=1}^{I} z_{i,m} \le 1, \forall m \in \mathcal{M}, & \text{(19e)} \\ \sum_{m=1}^{M} z_{i,m} \le 1, \forall i \in \mathcal{I}, & \text{(19f)} \\ z_{i,m} \in \{0,1\}, a_{i,m} \ge 0, \forall i \in \mathcal{I}, \forall m \in \mathcal{M}, & \text{(19g)} \end{cases}$$

where $d_{i,m,0}$ is the Euclidean distance between the spatial location required by task m and that of the charging station for UAV i, which can be obtained by

$$d_{i,m,0} = \sqrt{(X_{i,0} - X_m)^2 + (Y_{i,0} - Y_m)^2 + {H_m}^2}. \tag{20}$$

Here, $(X_{i,0}, Y_{i,0})$ is the horizontal location of the charging station for UAV i.

The input parameters of Problem 1 are the set of UAVs $\mathcal{I}$, the set of sensing tasks $\mathcal{M}$, each UAV's bid $\mathbf{b}_i$, the six-tuple of each sensing task Γ_m, the current spatial locations of UAVs, the estimated data size of sensing result $D_{i,m}$, and some other parameters including λ, γ, ω, and μ. The decision variables of Problem 1 are $\mathbf{Z}$ and compensating payment matrix $\mathbf{A} = [a_{i,m}]_{I\times M}$. Here, $z_{i,m} \in \{0,1\}$ is a binary variable. If $z_{i,m} = 1$, UAV i is selected to be the winner to perform sensing task m. Otherwise, $z_{i,m} = 0$. $a_{i,m}$ is the compensating payment to UAV i for performing sensing task m. As such, if UAV i is the winner to perform sensing task m, the payment to UAV i is $p_i = b_{i,m} + a_{i,m}$.

Besides, there exist seven constraints in Problem 1. Constraint (19a) means that each winning UAV should follow the energy constraint, i.e., its remaining energy should be larger than the energy consumption of performing sensing tasks and flying to the charging station. As such, the UAV that has no sufficient energy cannot be selected to perform tasks, i.e., $z_{i,m} = 0$. Constraint (19b) indicates the time constraints of winning UAVs, which guarantees that the sensing data should be returned to the sensing platform within the TTL of the sensing tasks. Constraint (19c) is the sensing quality requirement of the sensing task. Constraint (19d) means that only when equipped with the corresponding type of sensor, the UAV can be selected as the winner to perform a specific sensing task. In constraint (19d), $z_{i,m}\alpha_{i,\Phi_m} + z_{i,m} \in \{0,2\}$ means that only when equipped with the corresponding type of sensor, the UAV can be selected as the winner to perform a specific sensing task. To be specific, when $\alpha_{i,\Phi_m} = 0$, UAV i cannot be the winner to perform sensing task m, i.e., $z_{i,m}\alpha_{i,\Phi_m} + z_{i,m} = 0$. When $\alpha_{i,\Phi_m} = 1$, UAV i can be the winner to perform sensing task m, where $z_{i,m}\alpha_{i,\Phi_m} + z_{i,m}$ can be equal to 0 or 2. Constraint (19e) and constraint (19f) mean that each task is assigned to at most one UAV and each UAV can perform at most one task at the moment, respectively. Constraint (19g) indicates that $z_{i,m}$ is a binary variable. Apart from the above-mentioned constraints, Problem 1 should also satisfy the property constraint including *truthfulness*, and *individual rationality*, which are defined as follows.

Definition 1: BASIC is truthful, for UAV $i, i \in \mathcal{I}$ with $b_{i,m} \neq b^*_{i,m}$ on any sensing task $m, m \in \mathcal{M}$, and bids of other UAVs $\mathbf{b}_{-i,m}$, if there exists

$$U_{i,m}(b^*_{i,m},\ \mathbf{b}_{-i,m}) \ge U_{i,m}(b_{i,m},\ \mathbf{b}_{-i,m}), \tag{21}$$

where $b^*_{i,m} = C_{i,m}$ is the true bid of UAV i to perform sensing task m.

Definition 2: BASIC is individual rational if any UAV i has nonnegative utility to perform any sensing task m, i.e., $U_{i,m}(b^*_{i,m},\ \mathbf{b}_{-i,m}) \ge 0$.

According to Problem 1, the sensing platform should collect some information about each UAV, including location, remaining energy, flying velocity, transmission rate, the estimated data size of the sensing result, and task bid, etc. In this article, we consider information sensing with UAVs in the urban scene, where the wireless connection is continuously stable. In particular, although the number of UAVs is high and the connection of each UAV is dynamic, each UAV can submit its information to the closest base station via the ubiquitous wireless link in urban, such as 5G and 6G. Then, the sensing platform deployed in the cloud server with a connection to the internet can obtain information of UAVs. These information exchanges between the UAVs and the sensing platform inevitably consume their energy. However, in this article, each UAV does not upload its information all the time, and the information is submitted when the UAV participates in the task auction every certain time, which avoids the fast energy consumption of this UAV. In addition, when the remaining energy of the UAV becomes insufficient

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

to support the information sensing, the UAV will fly back to the charging station for recharging, whereby this UAV can be employed to conduct the sensing tasks for a long time. Besides, for the sensing platform, since it is deployed on the cloud server which has a constant energy supply, the sensing platform can keep online and assign sensing tasks at all time.

V. Solution and Analysis

In this section, we first present the solution to Problem 1 and then conduct the property analysis of BASIC.

A. Problem Solution

Although the problem of sensing task assignment in UAV-enabled sensing system is mathematically formulized as Problem 1, it is intractable due to its NP-hardness.

Theorem 1: Problem 1 is NP-hard.

Proof: First, the decision variable $z_{i,m}, \forall i \in \mathcal{I}$ and $\forall m \in \mathcal{M}$, is discrete and constraint (19g) is not convex. Besides, a special case is first considered. We relax the conditions of truthfulness, individual rationality, and constraints (19a)–(19d) in the context of Problem 1, while keeping the payment vector $\mathbf{p}$ constant. Here, Problem 1 can be seen as a weighted set cover problem, which has been proven to be NP-hard [51]. Therefore, Problem 1 is an NP-hard problem. This completes our proof. ■

Since Problem 1 is NP-hard, it is hard to directly search for the optimal sensing task assignment solution in polynomial time. As such, we propose a heuristic optimization algorithm to achieve the approximately optimal solution for Problem 1. The heuristic optimization algorithm includes two sub-algorithms. The first sub-algorithm is to select the optimal winning UAV matrix, i.e., $\mathbf{Z}^*$, and the second sub-algorithm is to determine the optimal payment, i.e., $\mathbf{p}^*$.

Winning UAV Selection: ($\mathbf{Z}^*$) The winning UAV selection sub-algorithm is shown in Algorithm 1, which is a modified greedy algorithm. Initially, for each sensing task m, the sensing platform builds a set $\mathcal{I}_m$ by ruling out UAVs who do not follow constraints (19a)–(19d) (steps 2-4). According to the utility of the sensing platform, each UAV can provide a certain utility increment to the sensing platform by performing a task. Here, we define a *utility contribution* for UAV i on sensing task m, which is expressed as

$$\Omega_{i,m}(b_{i,m}) = \lambda\Bigg\{-(1-\gamma)\left(\frac{d_{i,m}[t]}{v_i}+\frac{D_{i,m}}{R_i\left[t_{i,m}\right]}\right) + \gamma\varpi q_{i,\Phi_m}\Bigg\} - (1-\lambda)\omega b_{i,m}. \tag{22}$$

In each iteration, the sensing platform selects a winning UAV to perform a specific sensing task. Specifically, the sensing platform first calculates the maximum utility contribution for each sensing task m, i.e., $\max\{\Omega_{i,m}(b_{i,m})|i \in \mathcal{I}_m\}$ (step 6-15). Then, the sensing platform selects the winning UAV that provides the maximum utility contribution among all the maximum utility contributions for sensing tasks, i.e.,$\max_{i^*,m^*}\{\max_i\{\Omega_{i,m}(b_{i,m})\}|m \in \mathcal{M}'\}$ (step 16), where

Algorithm 1: Winning UAV Selection Sub-Algorithm.

Input: UAV set $\mathcal{I}$, task set $\mathcal{M}$, each UAV's bid $b_{i,m}$ ($\forall i \in \mathcal{I}, \forall m \in \mathcal{M}$), each UAV's sensing quality $q_{i,m}$ ($\forall i \in \mathcal{I}, \forall m \in \mathcal{M}$).
Output: Optimal winning UAV selection matrix $\mathbf{Z}^*$.

1 **Initialization:** $\mathbf{Z}^* = [0]_{I\times M}$, $\mathcal{M}' = \mathcal{M}$, $\mathcal{I}' = \mathcal{I}$;
2 **for** $m \in \mathcal{M}'$ **do**
3 Build set $\mathcal{I}_m$ from $\mathcal{I}'$ by ruling out UAVs that do not follow constraints (19a)-(19d);
4 **end**
5 **while** $\mathcal{M}' \neq \emptyset$ **do**
6 **for** $m \in \mathcal{M}'$ **do**
7 **if** $\mathcal{I}_m$ *is empty* **then**
8 $\mathcal{M}' = \mathcal{M}'\backslash\{m\}$;
9 **break**;
10 **end**
11 **for** $i \in \mathcal{I}_m$ **do**
12 Calculate the utility contribution $\Omega_{i,m}(b_{i,m})$ of UAV i on sensing task m by (22);
13 **end**
14 Calculate the maximum utility contribution for sensing task m, i.e., $\widetilde{\Omega}_{i',m}(b_{i',m}) = \max\{\Omega_{i,m}(b_{i,m})|i \in \mathcal{I}_m\}$;
15 **end**
16 Select the winner of a specific sensing task, where $(i^*, m^*) = \arg\max\{\widetilde{\Omega}_{i',m}(b_{i',m})|m \in \mathcal{M}\}$;
17 $z_{i^*,m^*} = 1$, $\mathcal{I}' = \mathcal{I}'\backslash\{i^*\}$ and $\mathcal{M}' = \mathcal{M}'\backslash\{m^*\}$;
18 **for** $m \in \mathcal{M}'$ **do**
19 Build set $\mathcal{I}_m$ from $\mathcal{I}'$ by ruling out UAVs who do not follow constraints (19a)-(19d);
20 **end**
21 **end**

$\mathcal{M}'$ is the set of sensing tasks that have not been assigned. The loop stops when all sensing tasks are assigned to UAVs or there is no appropriate UAV that meets the demands of sensing tasks, i.e., $\mathcal{M}'$ is empty or $\mathcal{I}_m, \forall m \in \mathcal{M}$ is empty. In Algorithm 1, the sensing tasks are allocated to UAVs at each time slot. For each sensing task m, the set of the candidate UAVs (i.e., $\mathcal{I}_m$) is first selected, which satisfy the constraints (19a)–(19d) of Problem 1. If $\mathcal{I}_m$ is empty, i.e., $\mathcal{I}_m = \emptyset$, there are no appropriate UAVs to perform task m, where task m cannot be completed at the current time slot. Then, task m can be allocated at the next time slot and can be completed by the UAV that newly joins the crowdsensing system and meets the demand of this task.

Payment Determination: ($\mathbf{p}^*$) After winning UAVs are selected, the optimal payments of the sensing platform to winners should be determined. To this end, the idea of critical payment is utilized, where each UAV wins the auction to perform the specific sensing task when the bid is lower than the critical payment, and loses the auction when its bid is higher than the critical payment.

The payment determination sub-algorithm is shown in Algorithm 2. The payment $\mathbf{p}$ is initially set to zero. The main loop then determines the payment to each winning UAV. For each winning UAV $i \in \{i|z_{i,m} = 1, \forall i \in \mathcal{I}, \forall m \in \mathcal{M}\}$, the winning

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2: Winning UAV Selection Sub-Algorithm.

Input: The winning UAV matrix $\mathbf{Z}^*$.
Output: The optimal payment vector $\mathbf{p}^*$.
Initialization: $\mathbf{p}^* = \mathbf{0}$;
for $i \in \{i|z_{i,m} = 1, \forall i \in \mathcal{I}, \forall m \in \mathcal{M}\}$ **do**
  $\widetilde{\mathcal{I}} = \mathcal{I} \backslash \{i\}$;
  Run winner selection procedure of Algorithm 1 with input of $\widetilde{\mathcal{I}}$ and obtain a new winning UAV matrix $\widetilde{\mathbf{Z}}^*$;
  Search the winning UAV j of task m from $\widetilde{\mathbf{Z}}^*$;
  Calculate the virtual bid $\widetilde{b}_{i,j,m}$ by (24);
  $p_i^* = \widetilde{b}_{i,j,m}$;
end

UAV selection sub-algorithm is run again with the input of the UAV set that excepts UAV i. Afterwards, a new winning UAV matrix $\widetilde{\mathbf{Z}}^*$ is obtained, which excludes UAV i. Virtual bid $\widetilde{b}_{i,j,m}$ is then calculated for UAV i to substitute UAV $j \in \{i|\widetilde{z}_{i,m} = 1, \forall i \in \mathcal{I}, \forall m \in \mathcal{M}\}$ who is selected as a winning UAV to perform sensing task m. This implies that

$$\Omega_{j,m}(b_{j,m}) = \Omega_{i,m}(\widetilde{b}_{i,j,m}). \tag{23}$$

Thus, $\widetilde{b}_{i,j,m}$ is calculated by

$$\begin{aligned}\tilde{b}_{i,j,m} &= \Big\{\lambda\left\{\gamma\varpi\left(q_{i,\Phi_m} - q_{j,\Phi_m}\right) - (1-\gamma)\Delta_{i,j,m}[t]\right\} \\ &\quad + (1-\lambda)\omega b_{j,m}\Big\}/\{(1-\lambda)\omega\},\end{aligned} \tag{24}$$

where

$$\Delta_{i,j,m}[t] = \frac{d_{i,m}[t]}{v_i} - \frac{d_{j,m}[t]}{v_j} + \frac{D_{i,m}}{R_i\left[t_{i,m}\right]} - \frac{D_{j,m}}{R_j\left[t_{j,m}\right]}. \tag{25}$$

Finally, the payment to UAV i is obtained by $p_i^* = \widetilde{b}_{i,j,m}$, which equals to its virtual bid.

The signaling complexities of Algorithms 1 and 2 are analyzed as follows, respectively. At the initial stage of Algorithm 1, each UAV uploads its information (e.g., locations, remaining energy, etc.) to the sensing platform. It needs I communications to complete the information uploading of UAVs. Then, the sensing platform sends the sensing task information including the sensing locations, sensing duration, and sensing quality to these I UAVs, which needs IM communications. Here, M is the number of the assigned sensing tasks. Afterward, each UAV submits the bid of each sensing task to the sensing platform, where the amount of communications is IM. At last, the winner of each sensing task is noticed by the sensing platform, which needs at most M communications. Therefore, according to [52], the signaling complexity of Algorithm 1 is $I + 2IM + M$. Algorithm 2 is utilized to determine the payments to the winners of the auction. The sensing platform determines the payments based on the bids and sensing capabilities of winning UAVs, which have been sent to the sensing platform by UAVs in Algorithm 1. As such, the sensing platform in Algorithm 2 only needs to send the payments to the winning UAVs, which needs at most M communications. Therefore, according to [52], the signaling complexity of Algorithm 2 is M.

The computational complexities of Algorithms 1 and 2 are analyzed as follows. In Algorithm 1, the set of candidate UAVs by ruling out UAVs that do not follow constraints (19a)-(19 d) in Problem 1 is first formed, where the number of iterations is M (step 2-4). Second, the main loop (step 5-21) terminates in worse case after M iterations. In each iteration, at most M sub-iterations are needed to select the winning UAVs for sensing tasks (step 6-20). In each sub-iterations, it takes $\mathcal{O}(I)$ time to calculate the utility contributions (i.e., $\Omega_{i,m}(b_{i,m})$) of UAVs (step11-13). Therefore, the computational complexity of Algorithm 1 is $\mathcal{O}(M + M^2 I)$. Compared with Algorithm 1, there is one more layer of loop in Algorithm 2 to determine the payments to winning UAVs, which executes for M iterations in worse case. Therefore, the computational complexity of Algorithm 2 is $\mathcal{O}(M^2 + M^3 I)$.

B. Property Analysis

Here, we analyze the properties that BASIC provides, including *truthfulness* and *individual rationality*.

Theorem 2: In BASIC, each UAV is truthful to submit a bid, i.e., $b_{i,m} = C_{i,m}, \forall i \in \mathcal{I}, \forall m \in \mathcal{M}$.

Proof: We assume that UAV i submits a bid $b_{i,m}$ that $b_{i,m} \neq C_{i,m}$. We first consider $b_{i,m} > C_{i,m}$, where there exist four cases.

- *Case 1:* UAV i wins with $b_{i,m}$ to perform sensing task m but loses with $C_{i,m}$. According to the utility contribution, we can have
$$\Omega_{i,m}\left(b_{i,m}\right) - \Omega_{i,m}\left(C_{i,m}\right) = (1-\lambda)\omega C_{i,m} - (1-\lambda)\omega b_{i,m} > 0. \tag{26}$$
As such, $C_{i,m} > b_{i,m}$, which is contradicted with the pre-assumption $b_{i,m} > C_{i,m}$. In other words, UAV i cannot win to perform sensing task with $b_{i,m}$ while losing with $C_{i,m}$, when $b_{i,m} > C_{i,m}$.
- *Case 2:* UAV i loses with $b_{i,m}$ but wins with $C_{i,m}$ to perform sensing task m. The utility of UAV i to perform sensing task m with bid $b_{i,m}$ is $U_{i,m}(b_{i,m}) = 0$, while that with bid $C_{i,m}$ is $U_{i,m}(C_{i,m}) = {p_i}^* - C_{i,m}$. According to the payment determination sub-algorithm, ${p_i}^* \geq C_{i,m}$, which induces $U_{i,m}(C_{i,m}) \geq 0$. As such, UAV i tends to announce bid $C_{i,m}$ for sensing task m.
- *Case 3:* UAV i loses on sensing task m with both $b_{i,m}$ and $C_{i,m}$. Then, $U_{i,m}(b_{i,m}) = U_{i,m}(C_{i,m}) = 0$.
- *Case 4:* UAV i wins with both $b_{i,m}$ and $C_{i,m}$ to perform sensing task m. According to Algorithm 2, UAV i receives the same payment p_i^* in both cases. As such, $U_{i,m}(b_{i,m}) = U_{i,m}(C_{i,m}) = p_i^* - C_{i,m}$.

Furthermore, we analyze the scenario $b_{i,m} < C_{i,m}$, which also has four cases.

- *Case 1:* UAV i wins with $b_{i,m}$ to execute sensing task m but loses with $C_{i,m}$. If UAV i wins with $b_{i,m}$, according to Algorithm 2, a new winning UAV set $\widetilde{\mathbf{Z}}^*$ is generated. The payment p_i^* to UAV i is the virtual bid that makes UAV i substitute UAV j, where $\Omega_{j,m}(b_{j,m}) = \Omega_{i,m}(p_i^*)$.

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

If UAV i loses to perform sensing task m with $C_{i,m}$, the winning UAV for sensing task m is same with $\widetilde{\mathbf{Z}}^*$. According to the winning UAV selection criterion and utility contribution, we can have $\Omega_{j,m}(b_{j,m}) > \Omega_{i,m}(C_{i,m})$. As such, $\Omega_{i,m}(C_{i,m}) < \Omega_{i,m}(p_i^*)$. Based on the definition of utility contribution in (22), we can obtain $p_i^* < C_{i,m}$. Then, the utility of UAV i with $b_{i,m}$ is negative, i.e., $U_{i,m}(b_{i,m}) = p_i^* - C_{i,m} < U_{i,m}(C_{i,m}) = 0$. Thereby, UAV i has no willingness to select payment $b_{i,m}$.

- *Case 2:* UAV i loses with $b_{i,m}$ to perform sensing task m but wins with $C_{i,m}$. Here, the utility of UAV i with $b_{i,m}$ is $U_{i,m}(b_{i,m}) = 0$, while that with $C_{i,m}$ is $U_{i,m}(C_{i,m}) = p_i^* - C_{i,m}$. According to the payment determination sub-algorithm, $U_{i,m}(C_{i,m}) = p_i^* - C_{i,m} \geq 0 = U_{i,m}(b_{i,m})$. Therefore, UAV i has more willingness to choose $C_{i,m}$ for sensing task m.
- *Case 3:* UAV i loses with both $b_{i,m}$ and $C_{i,m}$ to execute sensing task m. Then, $U_{i,m}(b_{i,m}) = U_{i,m}(C_{i,m}) = 0$.
- *Case 4:* UAV i wins with both $b_{i,m}$ and $C_{i,m}$ to execute sensing task m. As the payment is independent with the bid, $U_{i,m}(b_{i,m}) = U_{i,m}(C_{i,m}) = p_i^* - C_{i,m}$.

Overall, when the bid $b_{i,m} \neq C_{i,m}$ for UAV i to perform sensing task m, its utility with $C_{i,m}$ is larger than or equal to that with $b_{i,m}$. Therefore, each UAV tends to choose its cost as the bid and thus BASIC has the property of truthfulness. This completes our proof. ■

Theorem 3: UAVs obtain individual rationality in BASIC.

Proof: According to Theorem 2, each UAV submits a bid, $C_{i,m}, \forall i \in \mathcal{I}, \forall m \in \mathcal{M}$, to the sensing platform for each sensing task. For any UAV $i \in \mathcal{I}$, we consider two cases.

- *Case 1:* UAV i loses to perform sensing task m. Then, $U_{i,m}(C_{i,m}) = 0$.
- *Case 2:* UAV i wins to perform sensing task m. With the winning UAV selection sub-algorithm, if UAV i is selected as a winner to perform sensing task m, we can obtain $\Omega_{i,m}(C_{i,m}) \geq \Omega_{j,m}(C_{j,m}), \forall j \in \mathcal{I}_m, j \neq i$. To determine the payment to UAV i, a new winning UAV selection matrix $\widetilde{\mathbf{Z}}^*$ is generated by the payment determination sub-algorithm, where the input of algorithm 2 excludes UAV i. Here, assume UAV $j', \widetilde{z}^*_{j',m} = 1$ is the winner to perform sensing task m and thus we have $\Omega_{i,m}(p_i^*) = \Omega_{j',m}(C_{j',m})$. As UAV j' loses to perform the sensing task m when UAV i participates in the task assignment, it can be obtained that $\Omega_{i,m}(C_{i,m}) \geq \Omega_{j',m}(C_{j',m}) = \Omega_{i,m}(p_i{}^*)$. Then,

$$\Omega_{i,m}(C_{i,m}) - \Omega_{i,m}(p_i{}^*) = (1-\lambda)\omega p_i{}^* - (1-\lambda)\omega C_{i,m} \geq 0. \tag{27}$$

As such, $p_i{}^* \geq C_{i,m}$. The utility of UAV i to perform sensing task m is $U_{i,m}(b_{i,m}) = p_i{}^* - C_{i,m} \geq 0$. Therefore, all UAVs can obtain individual rationality in BASIC. ■

VI. Performance Evaluation

In this section, we carry out extensive simulations to evaluate the performance of the proposed BASIC. The simulation setup is first introduced, followed by the analysis and discussion of simulation results.

TABLE II
Simulation Parameters

Parameters	Notations	Values
Hovering power of UAV	P_i^h	$[40, 60]$ Watt
Sensing power of UAV	P_i^s	$[5, 10]$ Watt
Transmission power of UAV	P_i^t	$[1, 2]$ Watt
Transmission bandwidth of UAV	B_i	$[1, 2]$ MHz
Noise power spectrum density	σ_0^2	-170 dbm/Hz
Channel power gain with unit distance	β_0	-50 db
Path loss exponent of LoS link	μ	2
Positive flying power exponent	ν	1.8
Flying power conversion parameter	η_i	[0.7, 0.9]
Unit energy cost of UAV	c_i^e	0.01
Unit sensing cost of UAV	$c_{i,k}^s$	[0.8, 1.2]
Unit transmission cost of UAV	c_i^t	5
Weighted parameter of satisfaction	λ	0.6
Weighted parameter of sensing quality	γ	0.6
Adjusting coefficient of sensing quality	ϖ	300
Adjusting coefficient of payment	ω	0.05

A. Simulation Setup

In this article, we conduct extensive simulations to evaluate the performance of the proposed scheme. The simulation settings are based on works in [44], [46], [53], [54], which have been applied in the practical scenario. In the simulation, an 1000 m $\times$ 1000 m area is deployed to conduct experiments. The number of UAVs is randomly in $[10, 20]$ and the number of sensing tasks is randomly selected from $[10, 20]$. The number of sensing task types is 10. The quality demand of each sensing task follows the uniform distribution in $[0.1, 0.9]$. The TTL of each sensing task is randomly in $[30, 50]$ s. The sensing time duration of each sensing task is randomly in [5,10] s. The size of sensing data for each sensing task is selected from interval $[20, 50]$ Mb. The flying speed of each UAV follows the uniform distribution in $[10, 20]$ m/s. The flying altitude of each UAV is randomly selected from the uniform distribution $[1, 100]$ m. The maximum energy reverse of each UAV is in [25,35] Watt $\cdot$ h. The number of charging stations is 2, whose horizontal locations are [0,0] and [1000,1000], respectively. Each UAV belongs to an arbitrary charging station with a probability of 0.5. The deployment of BSs follows a Poisson point process with the mean value of 0.005. Other settings are given in Table II. The simulation time is set to be 200 s. The sensing tasks are generated and the statuses of UAVs are updated every 5 sec. The horizontal location and altitude required by each sensing task are randomly set in this area. Each simulation is repeated 100 times.

Based on the above conditions, we also compare BASIC with the following benchmark schemes to show its superiority.

Greedy-algorithm-based task assignment scheme (GATA) [55]: The sensing platform utilizes the greedy algorithm to select winning UAVs. Each sensing task is allocated to the current UAV that can provide the maximum utility contribution for the sensing platform. The payments to the winning UAVs are determined similarly to Algorithm 2, which is based on the critical payment.

Second-price-sealed-auction-based task assignment scheme (SPSA) [56]: The sensing platform selects winning UAVs based on the bids announced by UAVs, where a group of UAVs that have the minimum bids are chosen to perform sensing tasks.

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

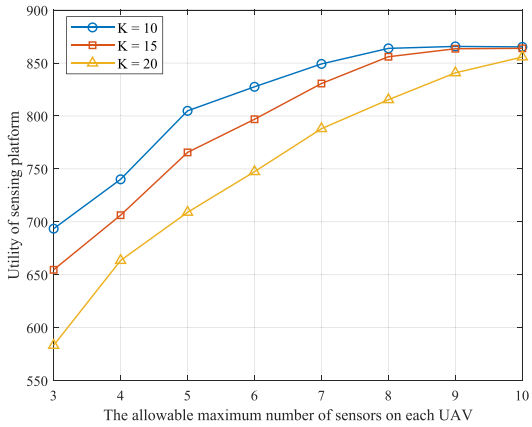


Fig. 3. Utility of the sensing platform vs. the allowable maximum number of sensors on each UAV.

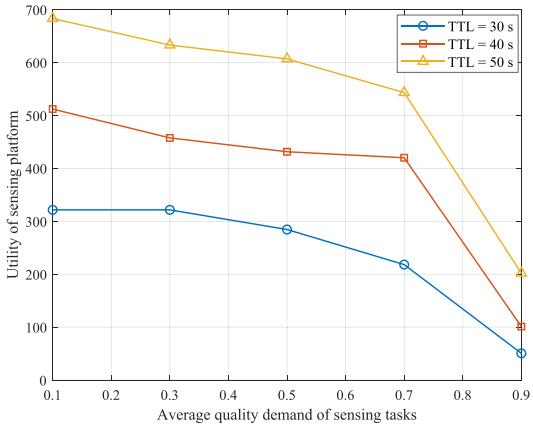


Fig. 4. Utility of the sensing platform vs. average quality demand of sensing tasks.

The bid of each UAV is the cost to execute the sensing tasks. The payment to the winning UAV of each sensing task is the second-lowest bid.

First-price-sealed-auction-based task assignment scheme (FPSA) [28]: Similar to SPSA, the sensing platform selects winning UAVs that have the minimum bids to perform sensing tasks. The bid of each UAV is based on the cost to execute the sensing tasks and the evaluation on costs of other UAVs. The payment to each winning UAV of each sensing task is the announced bid.

B. Simulation Results

Fig. 3 shows the utility of the sensing platform vs. the allowable maximum number of sensors on each UAV, i.e., Λ. Here, Λ changes from 3 to 10. The number of UAVs is set to 15 and the number of sensing tasks is set to 20. The number of sensing task types K varies among $\{10, 15, 20\}$. It can be first observed that the utility of the sensing platform increases with the Λ. This is because more UAVs can be selected to perform the sensing tasks when the allowable maximum number of sensors on each UAV becomes larger. Furthermore, the increasing rate decreases with the increase of Λ, and finally the utility of the sensing platform reaches to be stable. The reason is that all sensing tasks can be generally executed by UAVs if Λ is larger than a certain value. For example, all sensing tasks can be executed by UAVs if $\Lambda \geq 8$, when $K = 10$. Besides, when Λ is fixed, the utility of the sensing platform is large if the number of sensing task types is small. It can be explained as follows. When the number of sensing task types is smaller, more UAVs can be equipped with the sensors of corresponding types to perform sensing tasks, even if the allowable maximum number of sensors on each UAV is small.

Fig. 4 shows the utility of the sensing platform vs. the average quality demand of sensing tasks. Here, the average utility demand of sensing tasks changes from 0.1 to 0.9. The TTL of each sensing task varies from 30 s to 50 s. The number of sensing task types is set to 10 and the maximum allowable number of sensors on each UAV is set to 10. Other settings

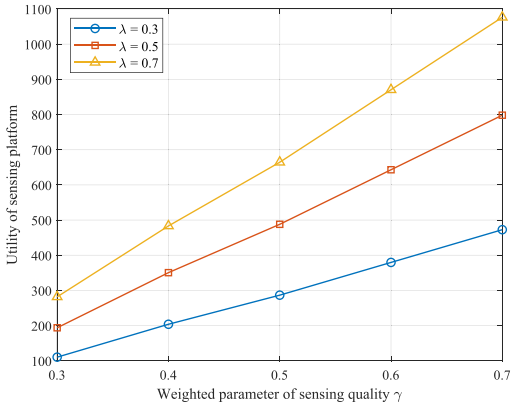


Fig. 5. Utility of the sensing platform vs. weighted parameter of sensing quality.

are unchanged. It can be seen that when TTL is fixed, the utility of the sensing platform decreases with the average demand of sensing tasks. The reason is that the number of appropriate UAVs to perform sensing tasks becomes smaller when the average demand of sensing tasks increases. As such, the sensing platform has to determine high payments to winning UAVs according to Algorithm 2. Especially, when the average quality demand of sensing tasks is large, there may be no UAVs that satisfy the quality demands of some sensing tasks, such that these sensing tasks cannot be executed. In addition, we can observe that when the average quality demand of sensing tasks is fixed, the utility of the sensing platform is large if the TTL is high. It is because more UAVs can act as the candidates to perform the sensing task when TTL is high, whereby the UAVs that provide larger utilities are selected as winners to execute sensing tasks.

Fig. 5 shows the utility of the sensing platform vs. the weighted parameter of sensing quality. Here, the weighted parameter of sensing quality, γ, changes from 0.3 to 0.7. The weighted parameter of satisfaction, λ, changes from 0.3 to 0.7. The TTL of each UAV randomly changes in $[30, 50]$ s, and the average quality demand of sensing tasks is set to 0.3. It can be observed that the utility of the sensing platform increases with the weighted parameter of sensing quality. For example,

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

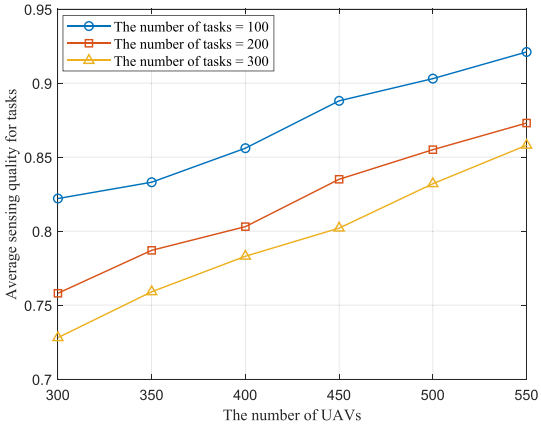
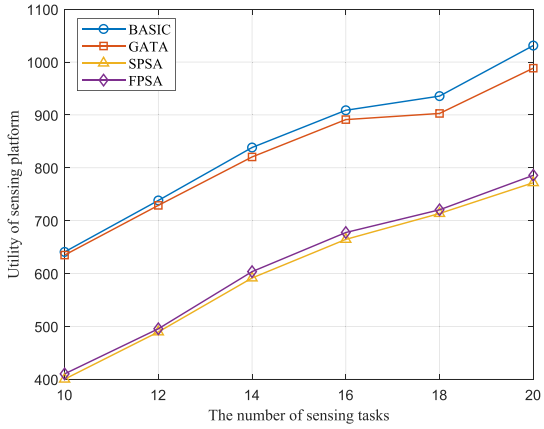
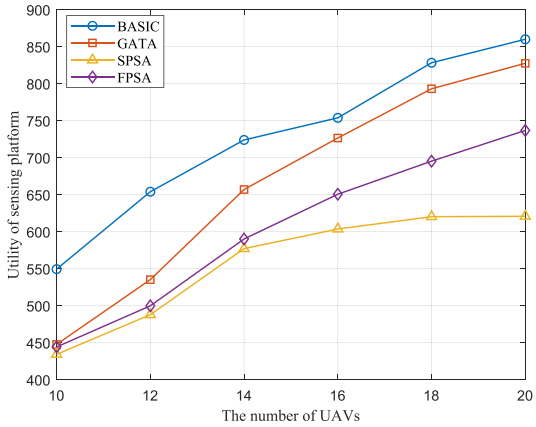


Fig. 6. Average sensing quality for tasks vs. the number of UAVs in a large scale crowdsensing system.



(a)



(b)

Fig. 7. Comparison of BASIC with benchmark schemes on the utility of sensing platform. (a) Comparison of four schemes, where the number of sensing tasks changes from 10 to 20. (b) Comparison of four schemes, where the number of sensing tasks changes from 10 to 20.

when $\lambda = 0.3$, the utility of sensing platform increases from 390 to 490, if γ changes from 0.6 to 0.7. This can be explained as follows. When the weighted parameter of sensing quality becomes larger, the sensing platform intends to select UAVs with higher sensing qualities, while the change of sensing time cost is nearly neglected. In addition, if γ is fixed, the utility of the sensing platform also increases with the weighted parameter of satisfaction. For example, when $\gamma = 0.5$, the utility of sensing platform increases from 490 to 670, where λ enlarges from 0.5 to 0.7. The reason is that the sensing platform selects UAVs that provide higher utility contributions when λ is larger, while the attention to the payments is decreased.

Fig. 6 shows the average sensing quality for tasks vs. the number of UAVs in a large-scale crowdsensing system. Here, the number of UAVs changes from 300 to 550, and the number of sensing tasks changes from 100 to 300. It can be observed that average sensing quality for tasks increases with the number of UAVs. For example, when the number of sensing tasks is 100, average sensing quality for tasks increases from 0.825 to 0.922, if the number of UAVs changes from 300 to 550. This can be explained as follows. Winners can be selected from more UAVs that provide high sensing qualities when the number of UAVs becomes larger. In addition, if the number of UAVs is fixed, average sensing quality for tasks decreases with the number of sensing tasks. For example, when the number of UAVs is 350, average sensing quality for tasks decreases from 0.842 to 0.761, where the number of sensing tasks enlarges from 100 to 300. This is because more sensing tasks have to be performed by UAVs with low sensing qualities, when the number of sensing tasks increases and the number of UAVs is unchanged.

Fig. 7 shows the comparison of BASIC with the other three conventional schemes on the utility of the sensing platform. Fig. 7(a) is the comparison on the utility of the sensing platform when the number of sensing tasks increases from 10 to 20. Fig. 7(b) is the comparison between four schemes, when the number of UAVs increases from 10 to 20. Here, we set $I = 20$, $\lambda = 0.6$, and $\gamma = 0.6$. Other settings are unchanged. From both Fig. 7(a) and (b), it is observed that the utility of the sensing platform in BASIC is the largest among four schemes. It can be explained as follows. In GATA, the sensing platform selects winning UAVs by calculating the maximum utility contribution for each sensing task, while the maximum utility contribution that each UAV can generate is not considered. As such, the winning UAV of a sensing task may provide a larger utility contribution when performing another sensing task. In SPSA, the sensing platform selects winning UAVs with the lowest bids, while UAVs that have low bids also provide low sensing qualities and have large sensing pending time lengths. In FPSA, UAVs with high sensing qualities can announce low bidding prices to be the winners. However, in BASIC, the sensing platform comprehensively considers the bid, sensing quality, and sensing pending time, where UAVs with the highest unit contributions are selected as winners to perform sensing tasks. In addition, from Fig. 7(a), the utility of the sensing platform increases with the number of sensing tasks. The reason is that more sensing tasks can be executed by UAVs when the number of sensing tasks increases. From Fig. 7(b), the utility of the sensing platform increases with the number of UAVs. This is because more appropriate UAVs act as candidates to perform sensing

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

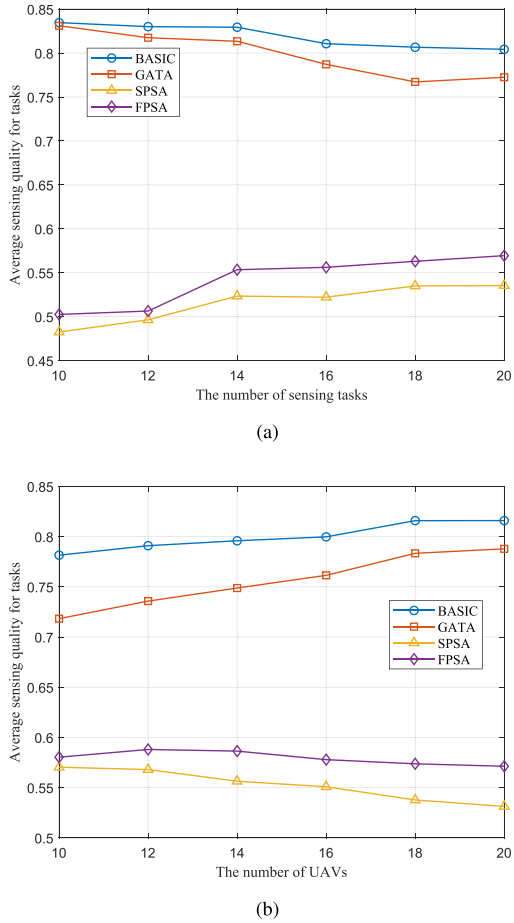


Fig. 8. Comparison of BASIC with benchmark schemes on average sensing quality for tasks. (a) Comparison of four schemes, where the number of sensing tasks changes from 10 to 20. (b) Comparison of four schemes, where the number of sensing tasks changes from 10 to 20.

tasks and provide high utility contributions when the number of UAVs becomes larger.

Fig. 8 shows the comparison of BASIC with the other three conventional schemes on the average sensing quality for tasks. Fig. 8(a) is the comparison on the average sensing quality for tasks when the number of sensing tasks increases from 10 to 20. Fig. 8(b) is the comparison on the average sensing quality for tasks, when the number of sensing tasks increases from 10 to 20. From both Fig. 8(a) and (b), it can be seen that the average sensing quality in BASIC is larger than those in GATA and SPSA. This can be explained as follows. In GATA, the UAV with the high sensing quality for a task is selected as a winner and ruled out from the candidates for other tasks, while this UAV may provide higher sensing quality for another task. In SPSA, the UAVs with the lowest bids are selected as winners to perform the sensing task, while these winning UAVs also provide low sensing qualities. In FPSA, UAVs with the low sensing quality may provide low bids, which can be selected as winners to perform sensing tasks. However, in BASIC, the sensing platform selects winners from UAVs that provide the highest sensing quality for all tasks. Besides, from Fig. 8(a), the average sensing quality for tasks in BASIC gradually decreases when the number of sensing tasks becomes larger. The reason is that the UAVs with low sensing qualities may be selected as winners if the number of sensing tasks is large. Additionally, from Fig. 8(b), the average sensing quality for tasks in BASIC gradually increases with the increase of the number of UAVs. This is because the sensing platform can select winners from more UAVs that provide high sensing qualities when the number of UAVs is larger. Besides, In SPSA, the sensing platform selects the UAV with the lowest bid as the winner to conduct sensing tasks. UAVs determine the bids based on the sensing cost, where the low sensing quality induces a low sensing cost. As such, the winning UAVs usually have low sensing quality. Especially, when the number of UAVs increases, more UAVs with low sensing quality are selected, resulting in decreasing the average sensing qualities for tasks.

VII. Conclusion

In this article, we have proposed a distributed sensing task assignment scheme, named BASIC, in the UAV-enabled crowdsensing system. Specifically, we have first developed a UAV-enabled crowdsensing framework, which includes task requesters, a sensing platform, and multiple distributed UAVs, where UAVs are concrete executors of sensing tasks. To motivate the participation of UAVs, we have devised a reverse-auction-based incentive mechanism to conduct sensing task assignment. Then, to resolve the utility maximization problem for the sensing platform in the incentive mechanism, we have designed a heuristic algorithm including a winning UAV selection sub-algorithm and payment determination sub-algorithm for the sensing platform to select optimal winning UAVs and payments. Finally, extensive simulation results have shown that BASIC has significant superiority on the utility of sensing platform and average sensing quality for tasks. In future work, we will further consider the different priority settings of sensing tasks and the cooperation of UAVs and mobile users on task sensing in crowdsensing system.

References

[1] B. Skorup and C. Haaland, "How drones can help fight the coronavirus," *Mercatus Center Res. Paper Series, Special Ed. Policy Brief*, 2020.
[2] M. Angurala, M. Bala, S. S. Bamber, R. Kaur, and P. Singh, "An Internet of Things assisted drone based approach to reduce rapid spread of COVID-19," *J. Saf. Sci. Resilience*, vol. 1, pp. 31–35, 2020.
[3] "UAV deployments in COVID-19 response: Early lessons learned in the US through the disaster relief program," 2020. [Online]. Available: https://enterprise-insights.dji.com/en/covid19-response-webinar
[4] L. Qian, Y. Wu, F. Jiang, N. Yu, W. Lu, and B. Lin, "NOMA assisted multi-task multi-access mobile edge computing via deep reinforcement learning for industrial Internet of Things," *IEEE Trans. Ind. Inform.*, vol. 17, no. 8, pp. 5688–5698, Aug. 2021.
[5] "Global Internet of Things (IoT) enterprise drone shipments from 2019 to 2023," Statista Res. Dept., Feb. 2020. [Online]. Available: https://www.statista.com/statistics/1079558/global-drone-enterprise-iot
[6] Y. Wang, Z. Su, N. Zhang, and A. Benslimane, "Learning in the air: Secure federated learning for UAV-assisted crowdsensing," *IEEE Trans. Netw. Sci. Eng.*, vol. 8, no. 2, pp. 1055–1069, Apr./ Jun. 2021.
[7] C. H. Liu, C. Piao, and J. Tang, "Energy-efficient UAV crowdsensing with multiple charging stations by deep learning," in *Proc. IEEE Conf. Comput. Commun.*, 2020, pp. 199–208.
[8] B. Wang, Y. Sun, D. Liu, H. M. Nguyen, and T. Q. Duong, "Social-aware UAV-assisted mobile crowd sensing in stochastic and dynamic environments for disaster relief networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 1, pp. 1070–1074, Jan. 2020.

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

[9] Z. Zhou et al., "When mobile crowd sensing meets UAV: Energy-efficient task assignment and route planning," *IEEE Trans. Commun.*, vol. 66, no. 11, pp. 5526–5538, Nov. 2018.

[10] Z. Zhou, J. Feng, B. Ai, and M. Guizani, "Energy-efficient mobile crowd sensing based on unmanned aerial vehicles," in *Proc. IEEE Glob. Commun. Conf.*, 2018, pp. 1–6.

[11] "Safety, security and communications solutions to prevent the spread of COVID-19," BOSCH, 2020. [Online]. Available: https://www.boschsecurity.com/us/en/news/rethink-the-magazine/covid-19-relevant-use-cases

[12] "Pandemic drone test flights are monitoring social distancing," CNET, 2020. [Online]. Available: https://www.cnet.com/news/pandemic-drone-test-flights-will-monitor-social-distancing

[13] J. Liu, Y. Yang, D. Li, X. Deng, S. Huang, and H. Liu, "An incentive mechanism based on behavioural economics in location-based crowdsensing considering an uneven distribution of participants," *IEEE Trans. Mobile Comput.*, vol. 21, no. 1, pp. 44–62, Jan. 2022.

[14] J. Wang, X. Feng, T. Xu, H. Ning, and T. Qiu, "Blockchain-based model for nondeterministic crowdsensing strategy with vehicular team cooperation," *IEEE Internet Things J.*, vol. 7, no. 9, pp. 8090–8098, Sep. 2020.

[15] C. Dai, X. Wang, K. Liu, D. Qi, W. Lin, and P. Zhou, "Stable task assignment for mobile crowdsensing with budget constraint," *IEEE Trans. Mobile Comput.*, vol. 20, no. 12, pp. 3439–3452, Dec. 2021.

[16] F. Wu, S. Yang, Z. Zheng, S. Tang, and G. Chen, "Fine-grained user profiling for personalized task matching in mobile crowdsensing," *IEEE Trans. Mobile Comput.*, vol. 20, no. 10, pp. 2961–2976, Oct. 2021.

[17] S. Xu, X. Chen, X. Pi, C. Joe-Wong, P. Zhang, and H. Y. Noh, "iLOCuS: Incentivizing vehicle mobility to optimize sensing distribution in crowd sensing," *IEEE Trans. Mobile Comput.*, vol. 19, no. 8, pp. 1831–1847, Aug. 2020.

[18] X. Niu, H. Huang, and Y. Li, "A real-time data collection mechanism with trajectory privacy in mobile crowd-sensing," *IEEE Commun. Lett.*, vol. 24, no. 10, pp. 2114–2118, Oct. 2020.

[19] G. Sun, L. Song, H. Yu, X. Du, and M. Guizani, "A two-tier collection and processing scheme for fog-based mobile crowd sensing in the internet of vehicles," *IEEE Internet Things J.*, vol. 8, no. 3, pp. 1971–1984, Feb. 2021.

[20] Q. Xu, Z. Su, M. Dai, and S. Yu, "APIS: Privacy-preserving incentive for sensing task allocation in cloud and edge-cooperation mobile Internet of Things with SDN," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 5892–5905, Jul. 2020.

[21] L. Zhang, Z. Li, and X. Chen, "Incentive mechanism design for edge-cloud collaboration in mobile crowd sensing," in *Proc. IEEE Conf. Comput. Commun. Workshops*, 2020, pp. 1196–1201.

[22] C. Zhu et al., "Folo: Latency and quality optimized task allocation in vehicular fog computing," *IEEE Internet Things J.*, vol. 6, no. 3, pp. 4150–4161, Jun. 2019.

[23] Q. Sun, L. Tian, Y. Zhou, J. Shi, and Z. Zhang, "Incentive scheme for slice cooperation based on D2D communication in 5G networks," *China Commun.*, vol. 17, no. 1, pp. 28–41, Jan. 2020.

[24] D. Zhao, H. Ma, and X. Ji, "Generalized lottery trees: Budget-balanced incentive tree mechanisms for crowdsourcing," *IEEE Trans. Mobile Comput.*, vol. 20, no. 7, pp. 2379–2394, Jul. 2021.

[25] J. Wang et al., "HyTasker: Hybrid task allocation in mobile crowd sensing," *IEEE Trans. Mobile Comput.*, vol. 19, no. 3, pp. 598–611, Mar. 2020.

[26] J. Xu, Z. Rao, L. Xu, D. Yang, and T. Li, "Incentive mechanism for multiple cooperative tasks with compatible users in mobile crowd sensing via online communities," *IEEE Trans. Mobile Comput.*, vol. 19, no. 7, pp. 1618–1633, Jul. 2020.

[27] X. Chen, Y. Zhao, Z. Li, and Y. Chen, "Incentive mechanism design based on stochastic game for multi-modality crowd sensing," in *Proc. IEEE Int. Conf. Smart Internet Things*, 2018, pp. 222–228.

[28] M. Dai, J. Li, Z. Su, W. Chen, Q. Xu, and S. Fu, "A privacy preservation based scheme for task assignment in Internet of Things," *IEEE Trans. Netw. Sci. Eng.*, vol. 7, no. 4, pp. 2323–2335, Oct.–Dec. 2020.

[29] Z. Dai, C. H. Liu, R. Han, G. Wang, K. Leung, and J. Tang, "Delay-sensitive energy-efficient UAV crowdsensing by deep reinforcement learning," *IEEE Trans. Mobile Comput.*, vol. 22, no. 4, pp. 2038–2052, Apr. 2023.

[30] K. Wei et al., "High-performance UAV crowdsensing: A deep reinforcement learning approach," *IEEE Internet Things J.*, vol. 9, no. 19, pp. 18487–18499, Oct. 2022.

[31] C. H. Liu, C. Piao, and J. Tang, "Energy-efficient UAV crowdsensing with multiple charging stations by deep learning," in *Proc. IEEE Conf. Comput. Commun.*, 2020, pp. 199–208.

[32] T. Cai et al., "Cooperative data sensing and computation offloading in UAV-assisted crowdsensing with multi-agent deep reinforcement learning," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 5, pp. 3197–3211, Sep./Oct. 2022.

[33] Y. Kang, S. Liu, H. Zhang, Z. Han, S. Osher, and H. V. Poor, "Task selection and collision-free route planning for mobile crowdsensing using multi-population mean-field games," *IEEE Trans. Green Commun. Netw.*, vol. 5, no. 4, pp. 1947–1960, Dec. 2021.

[34] W. Tan, L. Zhao, B. Li, L. Xu, and Y. Yang, "Multiple cooperative task allocation in group-oriented social mobile crowdsensing," *IEEE Trans. Serv. Comput.*, vol. 15, no. 6, pp. 3387–3401, Nov./Dec. 2022.

[35] X. Wang et al., "Online spatial crowdsensing with expertise-aware truth inference and task allocation," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 1, pp. 412–427, Jan. 2022.

[36] J.-J. Ji, Y.-N. Guo, X.-Z. Gao, D.-W. Gong, and Y.-P. Wang, "Q-learning-based hyperheuristic evolutionary algorithm for dynamic task allocation of crowdsensing," *IEEE Trans. Cybern.*, vol. 53, no. 4, pp. 2211–2224, Apr. 2023.

[37] Y. Qian, Y. Ma, J. Chen, D. Wu, D. Tian, and K. Hwang, "Optimal location privacy preserving and service quality guaranteed task allocation in vehicle-based crowdsensing networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 7, pp. 4367–4375, Jul. 2021.

[38] X. Li and X. Zhang, "Multi-task allocation under time constraints in mobile crowdsensing," *IEEE Trans. Mobile Comput.*, vol. 20, no. 4, pp. 1494–1510, Apr. 2021.

[39] Y. Li, F. Li, S. Yang, P. Zhou, L. Zhu, and Y. Wang, "Three-stage Stackelberg long-term incentive mechanism and monetization for mobile crowdsensing: An online learning approach," *IEEE Trans. Netw. Sci. Eng.*, vol. 8, no. 2, pp. 1385–1398, Apr./Jun. 2021.

[40] Y. Lin, Z. Cai, X. Wang, F. Hao, L. Wang, and A. M. V. V. Sai, "Multi-round incentive mechanism for cold start-enabled mobile crowdsensing," *IEEE Trans. Veh. Technol.*, vol. 70, no. 1, pp. 993–1007, Jan. 2021.

[41] J. Nie, J. Luo, Z. Xiong, D. Niyato, P. Wang, and H. V. Poor, "A multi-leader multi-follower game-based analysis for incentive mechanisms in socially-aware mobile crowdsensing," *IEEE Trans. Wireless Commun.*, vol. 20, no. 3, pp. 1457–1471, Mar. 2021.

[42] J. Xu, J. Xiang, and D. Yang, "Incentive mechanisms for time window dependent tasks in mobile crowdsensing," *IEEE Trans. Wireless Commun.*, vol. 14, no. 11, pp. 6353–6364, Nov. 2015.

[43] G. Ji, Z. Yao, B. Zhang, and C. Li, "Quality-driven online task-bundling-based incentive mechanism for mobile crowdsensing," *IEEE Trans. Veh. Technol.*, vol. 71, no. 7, pp. 7876–7889, Jul. 2022.

[44] H. Yan, Y. Chen, and S.-H. Yang, "UAV-enabled wireless power transfer with base station charging and UAV power consumption," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 12883–12896, Nov. 2020.

[45] Q. Wu, J. Xu, and R. Zhang, "Capacity characterization of UAV-enabled two-user broadcast channel," *IEEE J. Sel. Areas Commun.*, vol. 36, no. 9, pp. 1955–1971, Sep. 2018.

[46] F. Zhou, Y. Wu, R. Q. Hu, and Y. Qian, "Computation rate maximization in UAV-enabled wireless-powered mobile-edge computing systems," *IEEE J. Sel. Areas Commun.*, vol. 36, no. 9, pp. 1927–1941, Sep. 2018.

[47] T. Anttalainen, *Introduction to Telecommunications Network Engineering*. Norwood, MA, USA: Artech House, 2003.

[48] P. Klemperer, "Auction theory: A guide to the literature," *J. Econ. Surv.*, vol. 13, no. 3, pp. 227–286, 1999.

[49] R. Xing, Z. Su, T. H. Luan, Q. Xu, Y. Wang, and R. Li, "UAVs-aided delay-tolerant blockchain secure offline transactions in post-disaster vehicular networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 11, pp. 12030–12043, Nov. 2022.

[50] C. H. Liu, Z. Dai, H. Yang, and J. Tang, "Multi-task-oriented vehicular crowdsensing: A deep learning approach," in *Proc. IEEE Conf. Comput. Commun.*, 2020, pp. 1123–1132.

[51] M. Owais, M. K. Osman, and G. Moussa, "Multi-objective transit route network design as set covering problem," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 3, pp. 670–679, Mar. 2016.

[52] L. Cao, H. shun Tan, H. Peng, and M. cong Pan, "Multiple UAVs hierarchical dynamic task allocation based on PSO-FSA and decentralized auction," in *Proc. IEEE Int. Conf. Robot. Biomimetics*, 2014, pp. 2368–2373.

[53] Y. Wang, Z. Y. Ru, K. Wang, and P. Q. Huang, "Joint deployment and task scheduling optimization for large-scale mobile users in multi-UAV-enabled mobile edge computing," *IEEE Trans. Cybern.*, vol. 50, no. 9, pp. 3984–3997, Sep. 2020.

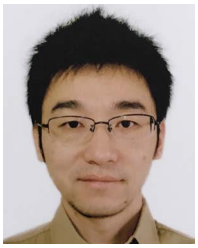
[54] Y. Zeng and R. Zhang, "Energy-efficient UAV communication with trajectory optimization," *IEEE Trans. Wireless Commun.*, vol. 16, no. 6, pp. 3747–3760, Jun. 2017.

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.

[55] X. Wang, W. Wu, and D. Qi, "Mobility-aware participant recruitment for vehicle-based mobile crowdsensing," *IEEE Trans. Veh. Technol.*, vol. 67, no. 5, pp. 4415–4426, May 2018.
[56] J. Zhang, M. Dai, and Z. Su, "Task allocation with unmanned surface vehicles in smart ocean IoT," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 9702–9713, Oct. 2020.



Qichao Xu received the Ph.D. degree from the School of Mechatronic Engineering and Automation, Shanghai University, Shanghai, China, in 2019. He is currently an Associate Professor with Shanghai University, Shanghai. He has authored or coauthored more than 50 papers in some respected journals, such as, IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY, IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING, IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, and IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY. His research interests include trust and security, the general area of wireless network architecture, Internet of Things, vehicular networks, and resource allocation. He was the recipient of the best paper awards from several international conferences, including IEEE IWCMC2022, IEEE MSN2020, EAI MONAMI2020, IEEE Comsoc GCCTC2018, IEEE CyberSciTech 2017, and WiCon2016.



Zhou Su (Senior Member, IEEE) has authored or coauthored technical papers, including top journals and top conferences, such as IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS, IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY, IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING, IEEE TRANSACTIONS ON MOBILE COMPUTING, IEEE/ACM TRANSACTIONS ON NETWORKING, and INFOCOM. His research interests include multimedia communication, wireless communication, and network traffic. He was the recipient of the Best Paper Award of International Conference IEEE ICC2020, IEEE BigdataSE2019, and IEEE CyberSciTech 2017. He is an Associate Editor for IEEE INTERNET OF THINGS JOURNAL, IEEE OPEN JOURNAL OF COMPUTER SOCIETY, and *IET Communications*.



Dongfeng Fang (Member, IEEE) received the B.S. degree in control theory and control engineering from the Harbin Institute of Technology, Harbin, China, in 2009, the M.S. degree in control theory and control engineering from Shanghai University, Shanghai, China, in 2013, and the Ph.D. degree in electrical and computer engineering from the University of Nebraska-Lincoln, Lincoln, NE, USA, in 2019. She is currently an Assistant Professor with the Department of Computer Science and Software Engineering, California Polytechnic State University, San Luis Obispo, CA, USA. Her research interests include cybersecurity (wireless security, cyber-physical security, critical infrastructure security, 5G security, IoT security, and privacy), wireless communications and networks, and public safety communications.



Yuan Wu received the Ph.D. degree in electronic and computer engineering from the Hong Kong University of Science and Technology, Hong Kong, in 2010. From 2016 to 2017, he was a Visiting Scholar with the Department of Electrical and Computer Engineering, University of Waterloo, Waterloo, On, Canada. He is currently an Associate Professor with the State Key Laboratory of Internet of Things for Smart City, University of Macau, Macau, China, and also with the Department of Computer and Information Science, University of Macau. His research interests include resource management for wireless networks, green communications and computing, mobile edge computing, and smart grids. He was the recipient of the Best Paper Award from the IEEE International Conference on Communications in 2016, Best Paper Award from IEEE Technical Committee on Green Communications and Computing in 2017, and Best Paper Award from the International Wireless Communications and Mobile Computing Conference in 2021. He was the Guest Editor of *IEEE Communications Magazine*, IEEE NETWORK, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, and *IET Communications*. He is on the Editorial Boards of IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, IEEE TRANSACTIONS ON NETWORK SCIENCE AND ENGINEERING, IEEE INTERNET OF THINGS JOURNAL, IEEE OPEN JOURNAL OF THE COMMUNICATIONS SOCIETY, and *China Communications*.

Authorized licensed use limited to: HUNAN UNIVERSITY. Downloaded on March 29,2024 at 06:55:51 UTC from IEEE Xplore. Restrictions apply.